USING THE TEXT OF FINNEGAN'S WAKE TO CREATE THE ELECTRO-ACOUSTIC WORK HUSH! CAUTION! ECHOLAND!

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ABSTRACT

Hush! Caution! Echoland! is part of a series of works entitled "Sounding Finnegan's Wake". Here computer/algorithmically generated instrumental, acousmatic, interactive, and audio/visual pieces are being developed, based on the sound and structure of Finnegan's Wake. The premise informing these works is that Finnegan's Wake can be understood as a textual work and as sound art. In Joyce's writing these two approaches are complementary, each informing and influencing each other.

Hush! Caution! Echoland! takes chapter one of Finnegan's Wake as its material, from which to create a score for traditional instruments and electronics, derived from the structure of the text as expressed initially through its ASCII representation. This is deconstructed through a number of processes to form the score for Flute, Clarinet, Violin, Viola and Cello. The electronic part also uses the ASCII representation of the text and spatializes the instrument sounds according to that structure. This is done through use of synthetic reverberation and panning techniques.

Processes for using text to generate music have been explored by many, including the author, and the outcomes are varied. These are discussed in the accompanying submission of the related creative work.

INTRODUCTION

This paper will discuss the processes used to generate Hush! Caution! Echoland!; its focus is on the developing of this work, with a brief discussion on precedents to it and its process.

Hush! Caution! Echoland! is part of a series of works entitled "Sounding Finnegan's Wake". In this series computer/algorithmically generated instrumental, acousmatic, interactive, and audio/visual pieces are being developed, based on the sound and structure of Finnegan's Wake. The premise informing these works is that Finnegan's Wake can be understood as a textual work and as sound art. In Joyce's writing these two approaches are complementary, each informing and influencing each other.

This part of "Sounding Finnegan's Wake" takes chapter one of Finnegan's Wake as its material, from which to create a score for traditional instruments and electronics of Hush! Caution! Echoland! The instrumental part is derived from the structure of the text as expressed initially through its ASCII representation. This is then deconstructed through a number of processes to form the score for Flute, Clarinet, Violin, Viola and Cello. The electronic part also uses the ASCII representation of the text and spatializes the instrument sounds according to that structure. This is done through use of synthetic reverberation and panning techniques.

Processes for using text to generate music have been explored by many, including the author, and the outcomes are varied. These processes range through an aesthetics based text setting, such as the writing of a pop song or liturgical music where the compose a response to what they they think complements the text or and answers the intentions of the author. Other approaches, such as using the inherent sounds of the text are comparatively rarer. Brunson's (Brunson 2009) discussion of processes is comprehensive, and Eberhard Blum's (Cage, Blum, and Cunningham 1991) renditions of Cage's Mesostics re Merce Cunningham and Amanda Stewart and Chris Mann's (Jones 2016) works are indicators of taking text from the descriptive realm to the less descriptive realm of sound art are examples. Processes for using text as a system for musical creation that is not designed to represent the meaning of the text are even rarer, the author has explored this (Alsop 1999, 2007, 2011, 2013, 2014), and this project is a continuation of these explorations. Work by others in the area is varied across a wide field ranging through exemplars and processes (Hammond and Reiner 2006, Manaris and Brown 2015, Kirke and Miranda 2017, Parson 2010, Rangarajan 2015, Harmon 2017, Laurie 2005).

There are also the less formal discussions and activities regarding Finnegans Wake and choices work in general, these include: Waywords and Meansigns (Pyle and Kipperman 2017), and From Swerve of Shore to Bend of Bay (Chrisp 2017). These contributions add significantly to the thinking and practice regarding Joyce and his works; it is their informality which creates a certain freedom that is not always available within the more academic contexts.

ASCII TO SCORE

While ASCII in itself is not a representation of word sounds, what it can do is indicate the relational structure of letters within a document, and these letters can represent, at a most basic level, word sounds. As this part of the Sounding Finnegans Wake Project is concerned with the structural underpinnings of Joyce's text, the use

of the ASCII values of each letter of the text presents an abstract process from which it is possible to develop a musical work.

The composition is loosely based on the following system, where 10 consecutive ASCII numbers create a single sonic event, the process is outlined below:

ASCII # 1 = pitch class
ASCII # 2 = register
ASCII # 3 = amplitude (dynamic ppp to fff)
ASCII # 4 = duration
ASCII # 5 = inter-onset time
ASCII # 6 = Instrument
ASCII # 7 = articulation
ASCII # 8 = virtual space (reverberation)
ASCII # 9 = actual space (panning)
ASCII # 10 = transition time

This process then repeats until all ASCII numbers, 12521 in total, in chapter one are used. This process doesn't consider whether the ASCII numbers are referring to actual letters, spaces, or punctuation marks. While the punctuation marks are not necessarily related to word sounds, they do indicate potential inflections or intonation characteristics, and the temporal aspects Joyce intended to imbue his work with.

Here, after a sequence of 10 ASCII numbers are iterated from the text, a new note is generated. Each note is then parsed by the composer to ensure fit with the chosen instrument. This system will be used to create a musical score for performance and instructions to the musician operating the electronics.

The number sequence to generate the first note is: 32 114 105 118 101 114 114 117 110 44. This then generates the outcome 56 48 30 1305 135 4 2 1583 6 356.

In this case the musical output can be seen in the A flat note played by the violin in measure one in **Figure 1**

ASCII # 1	32 (space)	pitch class	8
ASCII # 2	114 (r)	register	48
ASCII # 3	105 (i)	amplitude	30
ASCII # 4	118 (v)	Duration (msecs)	1305
ASCII # 5	101 (e)	inter-onset time (msece)	135
ASCII # 6	114 (r)	instrument	4
ASCII # 7	114 (r)	articulation	2
ASCII # 8	117 (u)	virtual space (reverberation)	1583
ASCII # 9	110 (n)	virtual space(panning)	6
ASCII # 10	44 (,)	transition time	356

Table 1. Conversion of 10 ASCII numbers to a note event

It should be noted that the resulting note value of 56 is addition of the original pitch class of 8 and the register value of 48.

Table 2 shows the selection of articulation choices made for this piece and **Table 3** shows the selection of dynamic ranges chosen. These choices are completely arbitrary and at the discretion of the composer with consideration of the capacity of each instrument.

	0	1	2	3
Flute	Trill (#)	Trill (b)	Vibrato	Flutter tongue
Clarinet	Trill (#)	Trill (b)	Vibrato	Flutter tongue
Violin	Trill (#)	Trill (b)	Battuto	Col Legno
Viola	Trill (#)	Trill (b)	Battuto	Col Legno
Cello	Trill (#)	Trill (b)	Battuto	Col Legno

Table 2. Variations in articulation.

MIDI velocity	Dynamic marking
0 – 15	ppp
15 – 30	pp
30 – 45	p
45 - 60	mp
65 – 80	mf
80 – 95	f
95 – 110	ff
110 - 127	fff

Table 3. Dynamic ranges.

A simplified version of the Max patch used to create this piece is shown in figures 1 and 2 below. **Figure 1** shows the pitch class, register, dynamic, duration, and inter-onset times for the note discussed earlier. After the designation of each column the top row shows the ASCII number, the second to fourth rows show the operations done on that number, and the final row shows the outcome that is then used to produce a note event.



Figure 1. creating a note event from ASCI numbers

To explain in greater detail: in column 1, which ascertains the note to be played, ASCII number 32, is fed through a modulo of 12, resulting in the number eight. This number is then added to the register value, which is ascertain in column two. Here the ASCII number is 114, which is being fed through modulo of four, as four possible register values are available, and the output is multiplied by 12 with a further edition of 24 to ensure that the last possible pitch is MIDI note C1. In this case their ASCII number is 114, which when put through a modulo of 4 results in two, which is then multiplied by 12 to create 24 after which 24 is added.

Similar processes are used to define the dynamic, duration, and inter-onset value, shown in **Figure 2**. Again it should be noted that the numbers chosen for the operations are done at the discretion of the composer.

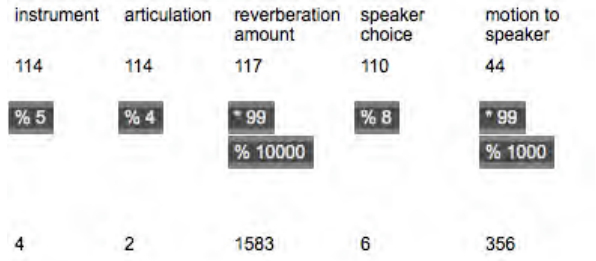


Figure 2. Defining instrument, articulation and electronic broadcast parameters.

Here the choice of instrument, in this case instrument 4, which was decided to be the violin. A modulo of five was used as 5 instruments were chosen, and an articulation 2, which case of the violin relates to using battuto, as shown in **Table 2.**

The reverberation amount is considered as the duration of the reverb tail in milliseconds, the speaker choice defines which speaker the amplifier to Sound of the specific instrument is to be played through, and the motion to speaker refers to the period of time taken for that specific instrument to move from the speaker and it was previously broadcast three to the current speaker.

DEVELOPING THE SCORE (INTERPRETING THE GENERATED DATA)

The score shown in figure 1 below is drafted in Logic Pro 10 from MIDI generated by Max patch used to create Hush! Caution! Echoland! This data has been quantised within 16th note triplets for performance, therefore the ordering of the notes on the page does not directly represent the ordering of notes as generated from the Max patch. In some cases the temporal positions of the notes as generated by the Max patch were in differences of milliseconds, and it was considered unlikely that the the audience would be able to hear these differences, musicians would be able to play the notes at such accuracy, or that the differences were vital to the musical ideas and expression. An example of this and be seen in the inter onset time of 135 milliseconds allocated to the first note as shown **Table 1.** Here it was considered that the real life interaction between musicians and the notes that they are playing would not show that particular duration.

This is considered appropriate, as any reading of a text is a sonic interpretation of it by the reader, just as any reading of score is a sonic interpretation of it by the musician. Other arbitrary decisions were made when generating the score. Occasionally note events that had very long durations were truncated, and pitches were either lowered or raised in order to fit within the range of chosen instrument; these choices were done at the composer's discretion.

As can be seen its note has a specific dynamic and specific articulation. These are designed as indicators to the performer, to be considered more as suggestions of way in which the note should be played. This approach harks to the way in which a script may be performed by an actor, where each individual actor is expected to bring their own interpretation of how the text should be spoken in performance.

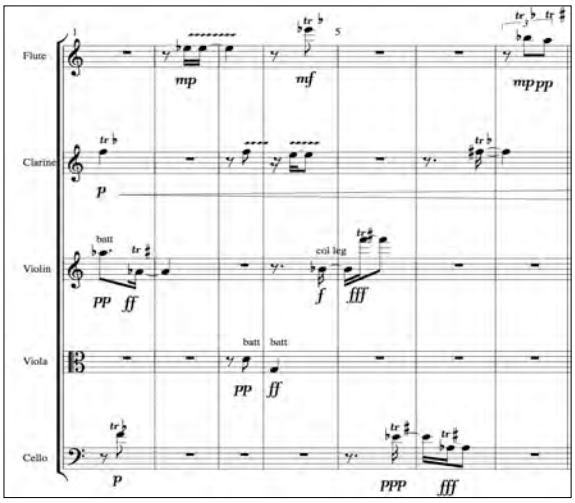


Figure 3. Example of the score to Hush! Caution! Echoland

CONCLUSION

This paper is a brief discussion of the processes used to develop a version of Finnegans Wake for acoustic instruments and virtual and actual space. It is part of a developing range of works that explore the possible sonic interpretations of this text. The process avoids traditional ideas of text setting, such as John Cage's "The Wonderful Widow of Eighteen Springs", or interpretive works such as Takemitsu's "riverrun". A comprehensive listing of musical settings and interpretations of Joyce's works is offered by Derek Pyle (2017). at Waywords and Meansigns.

The processes used in developing Hush! Caution! Echoland! will be developed in future works based on Finnegans Wake, in particular to develop works for solo instruments and voice.

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